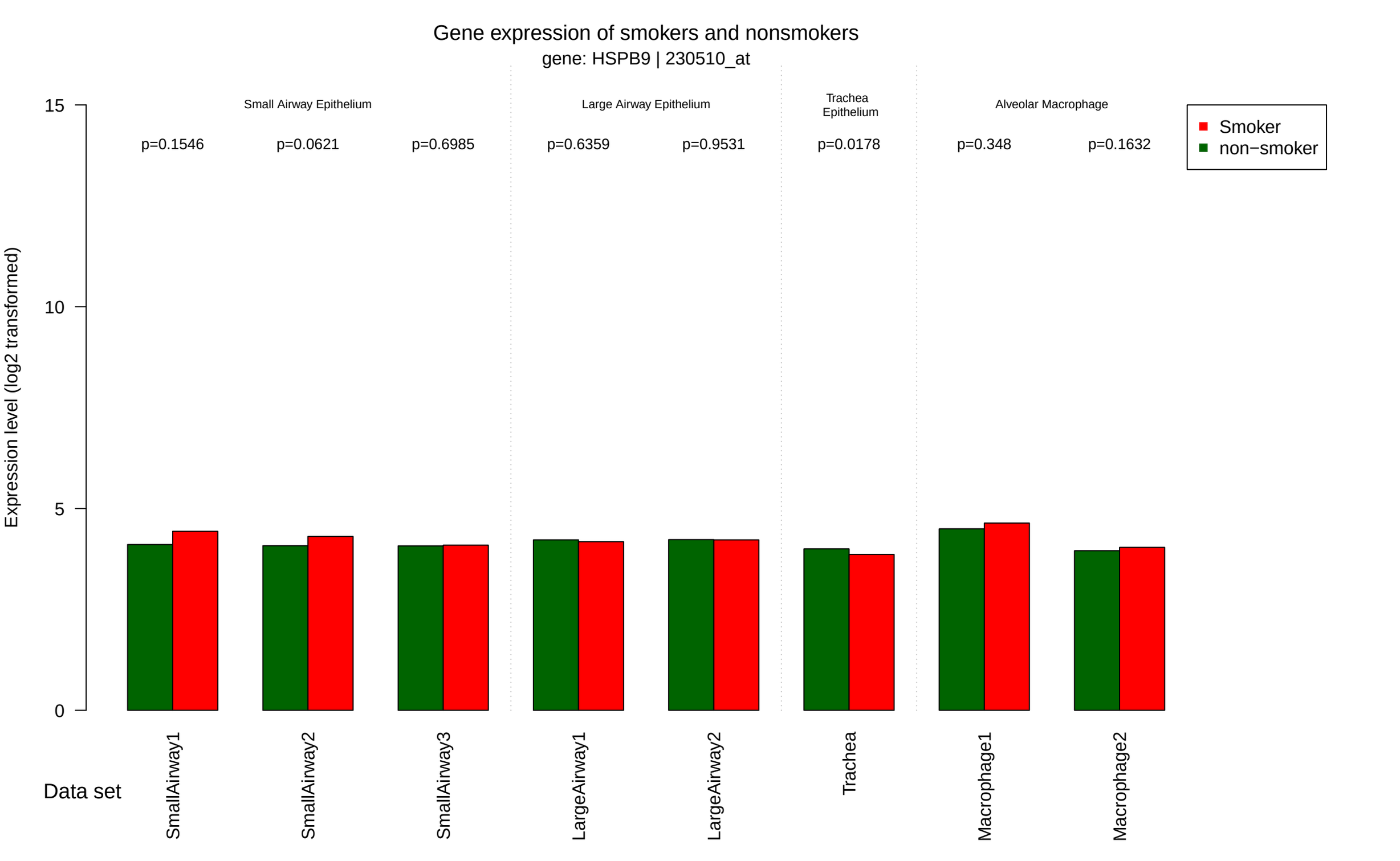
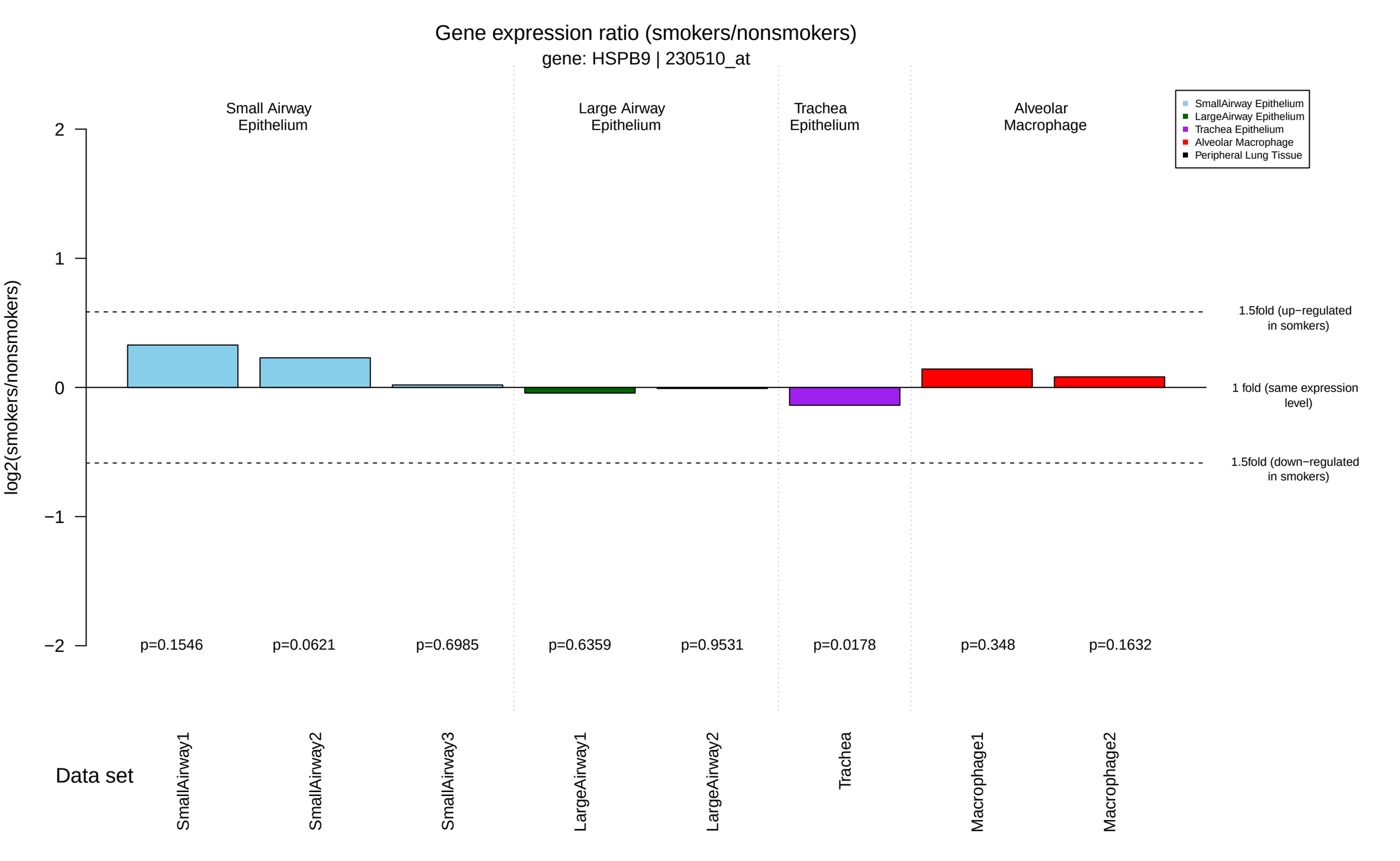
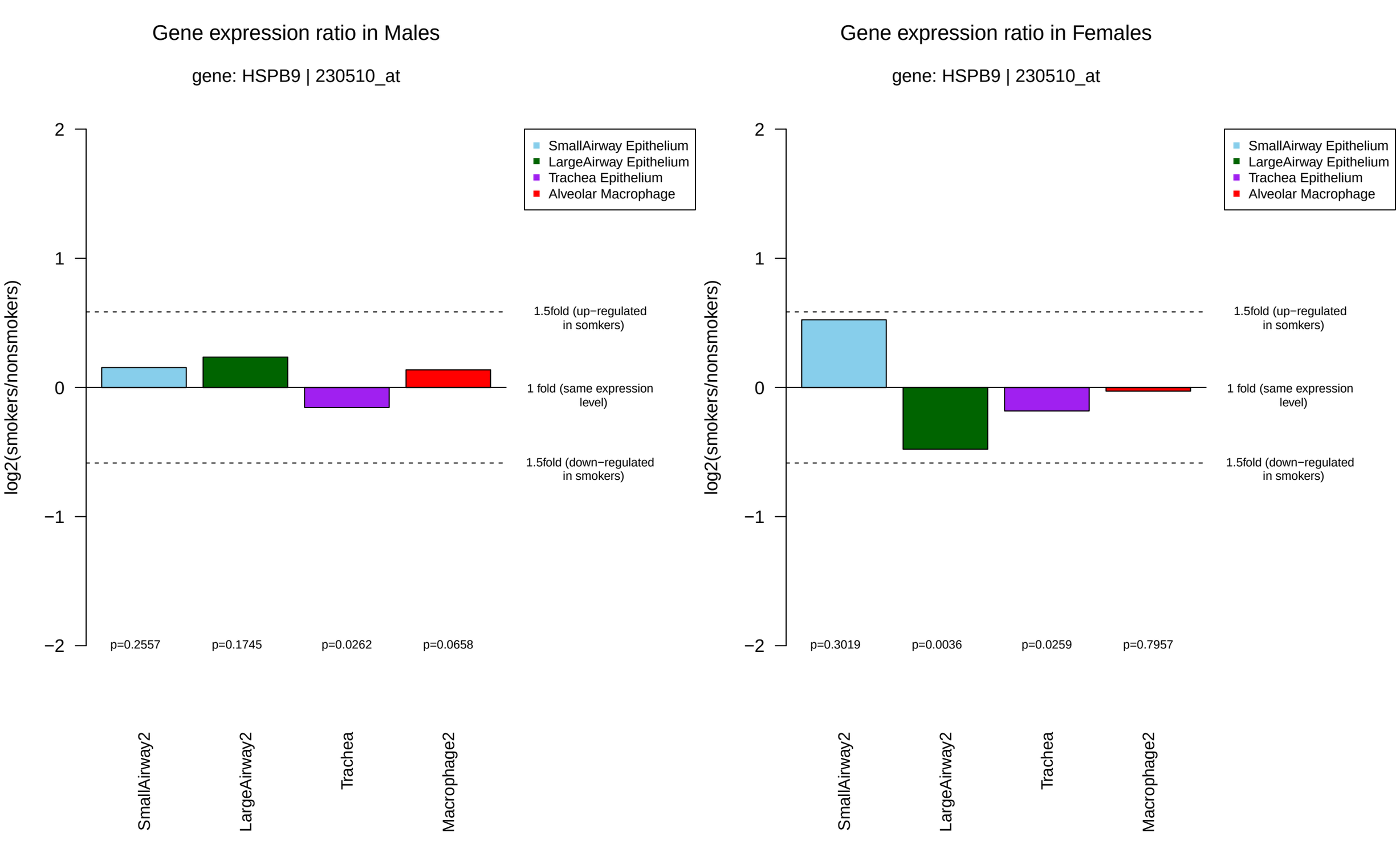
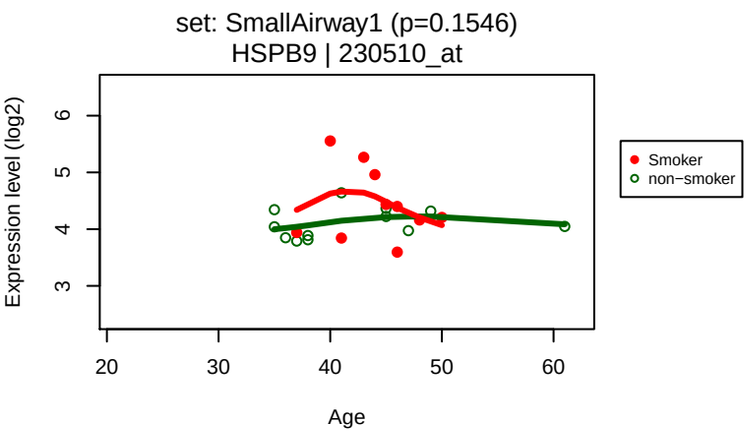
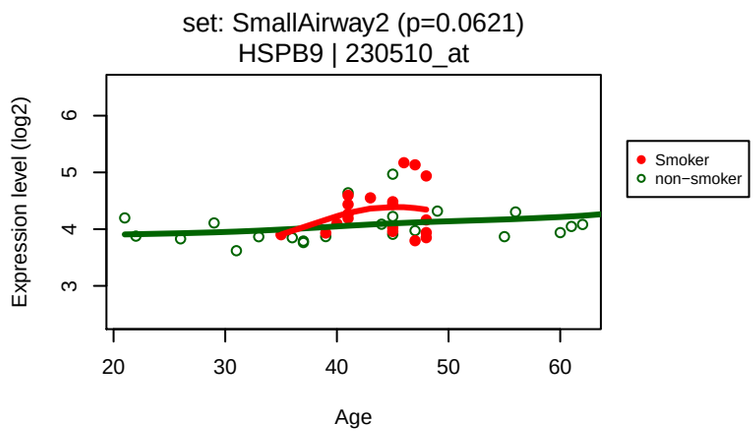
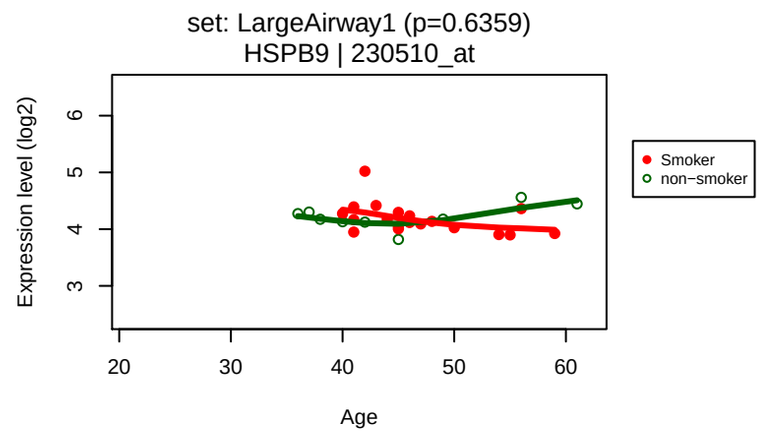

Gene expression of smokers and nonsmokers
gene: HSPB9 | 230510_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1546
p=0.0621
p=0.6985
p=0.6359
p=0.9531
p=0.0178
p=0.348
p=0.1632
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

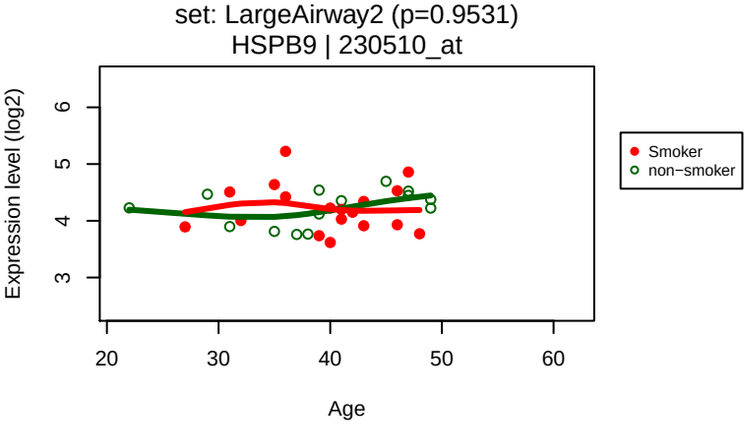

Gene expression ratio (smokers/nonsmokers)
gene: HSPB9 | 230510_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1546
p=0.0621
p=0.6985
p=0.6359
p=0.9531
p=0.0178
p=0.348
p=0.1632
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

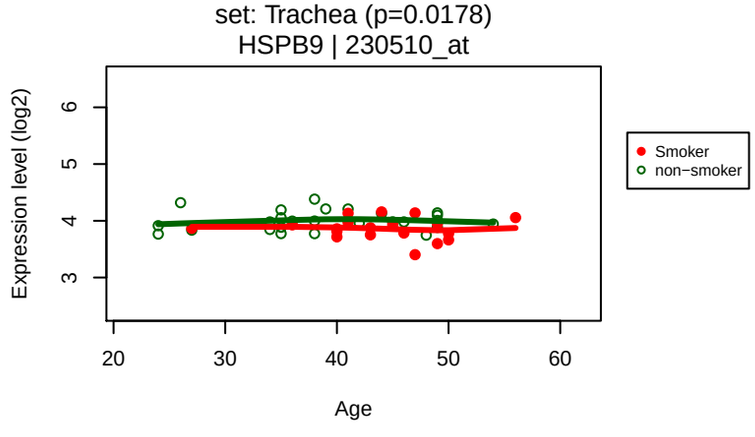

Gene expression ratio in Males
gene: HSPB9 | 230510_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2557
p=0.1745
p=0.0262
p=0.0658
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPB9 | 230510_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3019
p=0.0036
p=0.0259
p=0.7957
SmallAirway2
LargeAirway2
Trachea
Macrophage2

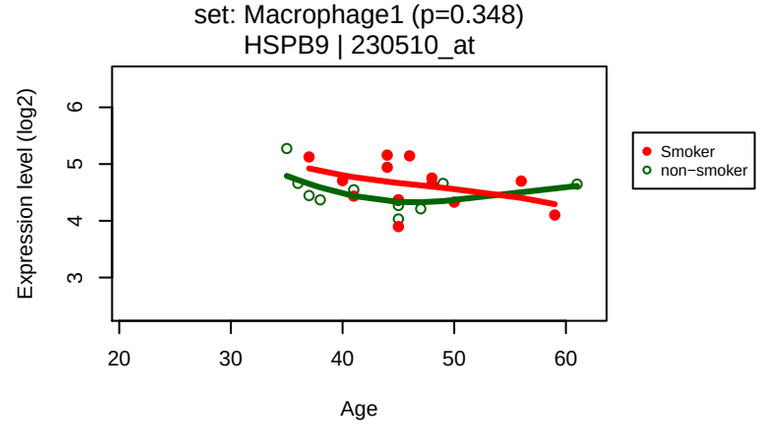

set: SmallAirway1 (p=0.1546)
HSPB9 | 230510_at
Expression level (log2)
Age
Smoker
non−smoker

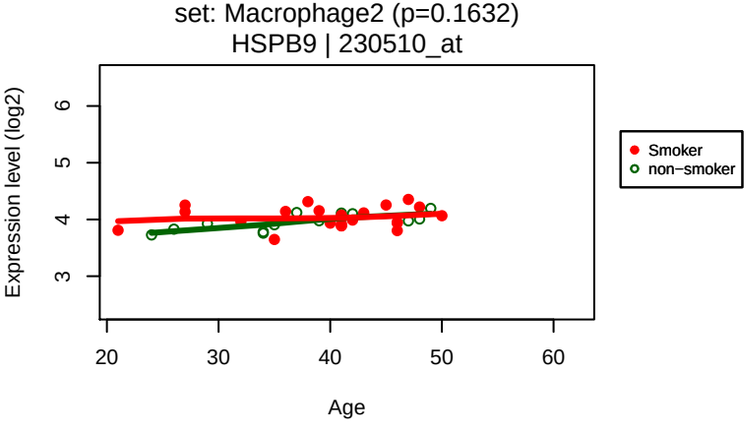

set: SmallAirway2 (p=0.0621)
HSPB9 | 230510_at
Expression level (log2)
Age
Smoker
non−smoker

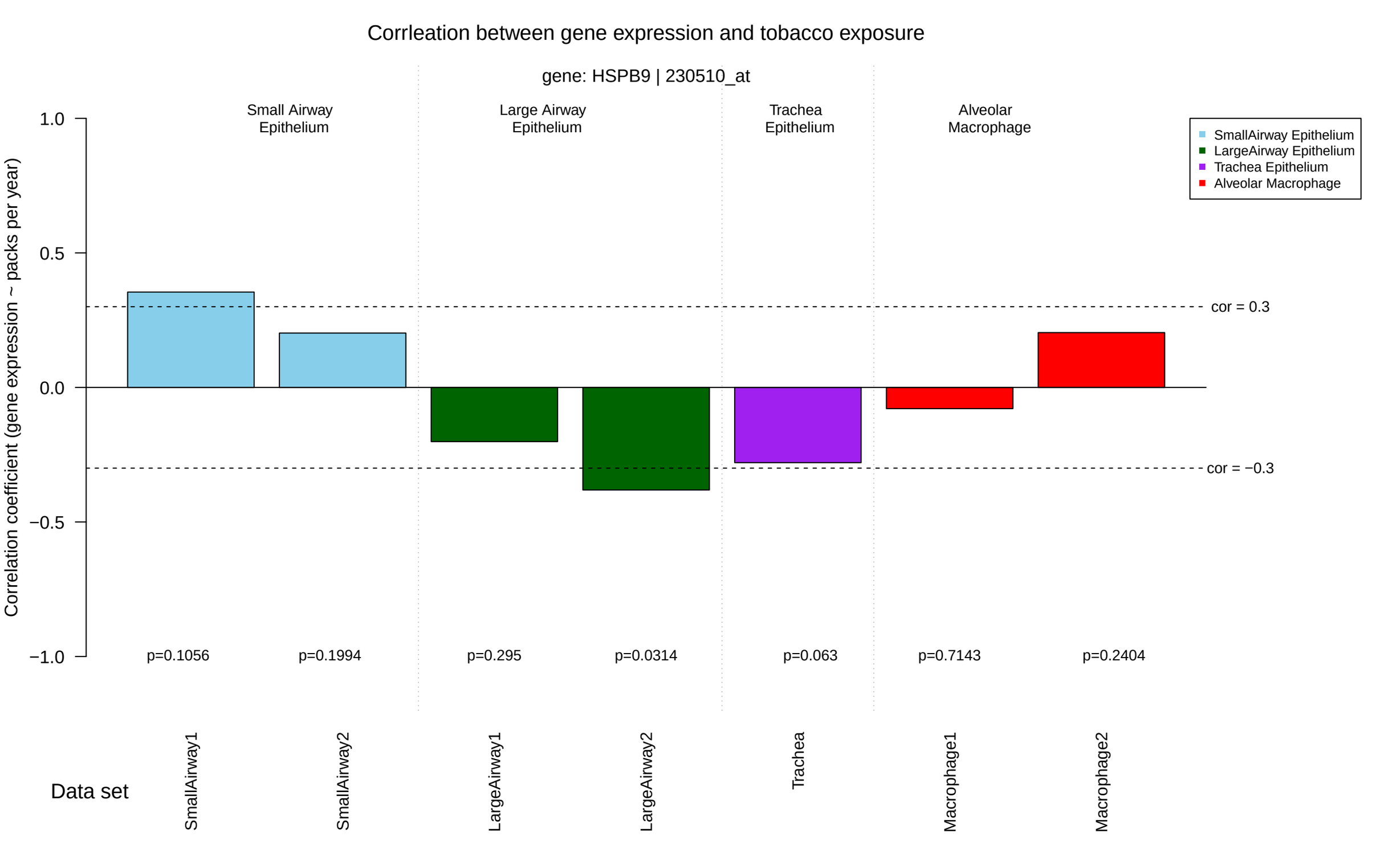

set: LargeAirway1 (p=0.6359)
HSPB9 | 230510_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9531)
HSPB9 | 230510_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0178)
HSPB9 | 230510_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.348)
HSPB9 | 230510_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1632)
HSPB9 | 230510_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPB9 | 230510_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1056
p=0.1994
p=0.295
p=0.0314
p=0.063
p=0.7143
p=0.2404
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2